# Answers and talk notes

Describe the Data Using the Median

### What your student is learning

Tonight we find the median when the data set has an EVEN number of values — so there is no single number sitting in the middle.

## Answers

Ask your student to explain before you look.

1. **B.** 6. Halfway between 4 and 8 is (4 + 8) ÷ 2 = 6.
2. **C.** 5. The middle numbers are 3 and 7: (3 + 7) ÷ 2 = 5.
3. **D.** 6. In order: 3, 5, 7, 9. The middle values are 5 and 7, so the median is (5 + 7) ÷ 2 = 6.
4. **A.** Yes, it can fall halfway between two values. Yes, it can fall halfway between two values, like the median of 2 and 5, which is 3.5.
5. **B.** 17. The middle values are 16 and 18: (16 + 18) ÷ 2 = 17.
6. **C.** 45 lb. In order: 38, 40, 44, 46, 52, 60. The median is (44 + 46) ÷ 2 = 45 lb.

### Talk about it

Why are there two middle numbers, and how did you find the number halfway between them?

### Home mission

Write 52, 55, 58, 62, 67 and 70 on six small papers and mix them up. Ask your student to put them in order and touch the two middle ones. Then ask: what number is halfway between them? Is there a paper with that number?

### When they are stuck, ask

- Is the list in order from least to greatest yet?
- How many values are there — odd or even?
- Which two numbers share the middle, and what is halfway between them?

### Try not to say

- Just take the number in the middle of the list as it was written.
- The median must be one of the numbers in the data.
- With an even count, pick either middle number, it hardly matters.